

Lab Goal : This lab was designed to teach you how to use recursion to solve a connection problem.

Lab Description : Take a group of provided connections and create a map of all connections. Search this map with two provided values to see if a connection exists. All connections are bi-directional. You will use a HashMap and a HashSet when solving this problem.

You can solve this with recursion using the DFS approach.
You can also solve this using a BFS approach with a queue.

If A is connected to B and B is connected to C, is A connected to C? yes

Sample Data :

```
9
CA XY RS YS ST TB AX BD RJ
CD
PQ QX AX BX CX DX EX FX GX AB BC CD DE AE CE FD TG
PT
AE EI IO OU BC CD DF FG
AG
HI HJ HK KL KM KN MO MP MQ
HQ
AB CD EF GH CB ED GF HI
AI
TV XY AZ XT JK KL LT JX MN TN JL NO OP PT NX
VZ
AB BC CD DE EF FG GH HI IJ JA AC FZ
AZ
NO PQ RS TU OU RP AB CD EF GH AH CE NS FA GQ
DT
IX VX CX DX MX LX BY
IB
```

Files Needed ::

```
//DFS
Graph.java
GraphRunner.java

//BFS
GraphQ.java
GraphQRunner.java

graph1.dat
```

Sample Output :

```
C CONNECTS TO D == CONNECTS
P CONNECTS TO T == CONNECTS
A CONNECTS TO G == DOES NOT CONNECT
H CONNECTS TO Q == CONNECTS
A CONNECTS TO I == CONNECTS
V CONNECTS TO Z == DOES NOT CONNECT
A CONNECTS TO Z == CONNECTS
D CONNECTS TO T == CONNECTS
I CONNECTS TO B == DOES NOT CONNECT
```